An Introduction to Quantum Computing

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Abstract. These lecture notes introduce quantum computing to graduate students in mathematics with a background in numerical linear algebra and numerical analysis. The central theme is *Quantum Phase Estimation* (QPE): every major algorithm — Shor's factoring algorithm, Grover search and amplitude estimation, the HHL quantum linear solver, and continuous-time quantum walks — is an instance of QPE applied to a carefully chosen unitary, with Stone's theorem connecting each unitary to its Hermitian generator. We develop the necessary foundations (quantum states, gates, measurement), the Quantum Fourier Transform as the computational engine, and QPE as a quantum eigenvalue algorithm. We then treat Shor's algorithm via the eigenstructure of modular exponentiation, Grover's algorithm and amplitude estimation, Hamiltonian simulation via product formulas, HHL, and continuous-time quantum walks. The course concludes with a survey of the quantum advantage landscape — distinguishing unconditional, hardness-conditional, and input-model-conditional speedups — a discussion of dequantization, and four open problems at the interface of quantum computing and numerical analysis. Throughout, connections to classical numerical methods are emphasized.

Contents

References

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1 Foundations

Quantum computing is, at its core, linear algebra over complex vector spaces, augmented by the physical constraints of quantum mechanics. We present the mathematical framework as four axioms, requiring no background in physics.

Why should we care? The central promise of quantum computing is computational speedup for problems that are intractable classically. For numerical mathematicians, the most relevant examples are: solving linear systems $Ax = b$ exponentially faster in the system size N; computing matrix functions such as e^A efficiently for large sparse matrices A; and estimating integrals and expectations with a quadratic improvement over Monte Carlo. These speedups come with important caveats — they require a quantum computer that does not yet exist at scale, and the results are often quantum states rather than classical vectors. Nevertheless, the algorithms are built from ideas that are deeply familiar to classical numerical methods: eigenvalue methods, Fourier analysis, operator splitting, and rational approximation. These notes try to develop those connections throughout.

The four postulates

State space. The *state* of a quantum system is described by a unit vector $|\psi\rangle$ in an N-dimensional complex Hilbert space $\mathcal{H}$. For n *qubits*, $\mathcal{H} = (\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}$.

Evolution. Closed-system evolution is described by a *unitary operator* $U \in \mathscr{U}(\mathcal{H})$. The state $|\psi\rangle$ becomes $U|\psi\rangle$ after applying the unitary U.

Measurement. A *projective measurement* with respect to an orthonormal basis $\{|k\rangle\}_{k=0}^{N-1}$ on state $|\psi\rangle$ yields outcome k with probability

$$p_k = |\langle k \,|\, \psi \rangle|^2, \qquad \text{(Born's rule)}$$

and leaves the post-measurement state $|k\rangle$.

Composition. The joint state space of systems A and B is $\mathcal{H}_A \otimes \mathcal{H}_B$. A gate U_A acting on A alone acts as $U_A \otimes \mathbb{I}_B$ on the joint system.

Remark 1.1. Each postulate has a direct interpretation in numerical linear algebra. A quantum state is a unit vector in $\mathbb{C}^{2^n}$, whose dimension grows exponentially in n. Gates are unitary matrices, so they are norm-preserving. The Born rule $p_k = |\langle k \,|\, \psi\rangle|^2$ is the squared magnitude of a projection coefficient in an orthonormal basis expansion. The tensor product $\mathbb{C}^{2^m} \otimes \mathbb{C}^{2^n} \cong \mathbb{C}^{2^{m+n}}$ is the standard multilinear tensor product.

Two states differing only by a *global phase*, $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ with $\gamma \in \mathbb{R}$, are indistinguishable: for every measurement basis $\{|k\rangle\}$ the Born probabilities $|\langle k \,|\, e^{i\gamma}\psi\rangle|^2 = |\langle k \,|\, \psi\rangle|^2$ coincide, and this persists after any later unitary (which commutes with the scalar $e^{i\gamma}$).

Using this freedom to make the $|0\rangle$-amplitude real and non-negative, every single-qubit state can be written

$$|\psi\rangle = \cos\tfrac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\tfrac{\theta}{2}\,|1\rangle, \qquad \theta \in [0, \pi],\ \phi \in [0, 2\pi).$$

The angles (θ, ϕ) are the spherical coordinates of a point on the unit sphere $S^2 \subset \mathbb{R}^3$, the *Bloch sphere*: $|0\rangle, |1\rangle$ are the north and south poles, while $|\pm\rangle$ and $|\pm y\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$ lie on the equator. Antipodal points are precisely the orthogonal states. Single-qubit unitaries act as rigid rotations of this sphere: $\mathsf{R}_x(\alpha)$, $\mathsf{R}_y(\alpha)$, $\mathsf{R}_z(\alpha)$ rotate by α about the $\hat{x}$, $\hat{y}$, $\hat{z}$ axes, and every $U \in \mathscr{U}(2)$ is, up to global phase, such a rotation (Euler decomposition $U = e^{i\gamma}\mathsf{R}_z(\beta)\mathsf{R}_y(\delta)\mathsf{R}_z(\lambda)$).

Exercise 1.2. Let $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle \in \mathbb{C}^2$ be an arbitrary single-qubit state.

(a) Compute the probability of outcome $|0\rangle$ when measuring in the computational basis.

(b) Compute the probability of outcome $|+\rangle$ when measuring in the basis $\{|+\rangle, |-\rangle\}$, where

$$|\pm\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle \pm |1\rangle\big).$$

(c) For which θ and ϕ are the two outcomes equally likely in both bases simultaneously?

Qubits and Dirac notation

A *qubit* is a unit vector in $\mathbb{C}^2$. The *computational basis* is $|0\rangle = (1,0)^\top$ and $|1\rangle = (0,1)^\top$, so a general qubit state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$, $\alpha, \beta \in \mathbb{C}$.

For n qubits, the computational basis is $\{|x\rangle : x \in \{0,1\}^n\}$, where $|x_1 \cdots x_n\rangle = |x_1\rangle \otimes \cdots \otimes |x_n\rangle$. A general n-qubit state is $|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle$ with $\sum_x |\alpha_x|^2 = 1$.

We use Dirac notation throughout. For $|\psi\rangle, |\phi\rangle \in \mathcal{H}$:

- $\langle\psi| = |\psi\rangle^\dagger$ is the conjugate transpose (row vector);
- $\langle\phi\,|\,\psi\rangle \in \mathbb{C}$ is the inner product;
- $|\phi\rangle\langle\psi|$ is the rank-1 operator (outer product);
- $\mathbb{I} = \sum_k |k\rangle\langle k|$ is the identity (resolution of the identity).

Standard quantum gates

Quantum gates are unitary matrices. The single-qubit *Pauli gates* are

$$\mathsf{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathsf{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \mathsf{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The *rotation gates* around the three Pauli axes are defined by

$$\mathsf{R}_x(\theta) = e^{-i(\theta/2)\mathsf{X}} = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix},$$

$$\mathsf{R}_y(\theta) = e^{-i(\theta/2)\mathsf{Y}} = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \qquad \mathsf{R}_z(\theta) = e^{-i(\theta/2)\mathsf{Z}} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

The *Hadamard gate* and *phase gates* are

$$\mathsf{H} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \mathsf{S} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \qquad \mathsf{T} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}, \qquad \mathsf{R}_\varphi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix}.$$

Note that $\mathsf{P} \in \{\mathsf{X}, \mathsf{Y}, \mathsf{Z}, \mathsf{H}\}$ are Hermitian ($\mathsf{P}^\dagger = \mathsf{P}$).

The canonical two-qubit gate is the *controlled-NOT*:

$$\mathsf{CNOT} = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes \mathsf{X} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad \mathsf{CNOT}|c\rangle|t\rangle = |c\rangle|c \oplus t\rangle.$$

More generally, a *controlled-U* gate applies U to the target qubit iff the control qubit is $|1\rangle$:

$$\text{c-}U = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes U.$$

Exercise 1.3. (a) Determine the eigenvalues and eigenvectors of X and Z.

(b) Show that H is the change-of-basis matrix between the eigenbases of Z and X, and verify the identity $\mathsf{HZH} = \mathsf{X}$ from this.

Exercise 1.4. Show that each Pauli gate satisfies $\mathsf{P}^2 = \mathbb{I}$ for $\mathsf{P} \in \{\mathsf{X}, \mathsf{Y}, \mathsf{Z}, \mathsf{H}\}$. Conclude from this that P takes the form $\mathsf{P} = e^{i(\pi/2)(\mathbb{I}-\mathsf{P})}$.

The Hadamard transform

Proposition 1.5 (Hadamard transform). *For any $x \in \{0,1\}^n$,*

$$\mathsf{H}^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z\in\{0,1\}^n} (-1)^{x\cdot z}|z\rangle,$$

where $x \cdot z = \sum_i x_i z_i \bmod 2$. In particular, $\mathsf{H}^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{2^n}} \sum_z |z\rangle$ is the uniform superposition over all basis states.

Proof. For $n = 1$: $\mathsf{H}|x_1\rangle = \frac{1}{\sqrt{2}} \sum_{z_1} (-1)^{x_1 z_1}|z_1\rangle$. For general n, apply the base case qubit-by-qubit and collect the signs over the tensor product:

$$\mathsf{H}^{\otimes n}|x\rangle = \bigotimes_{i=1}^{n} \mathsf{H}|x_i\rangle = \bigotimes_{i=1}^{n} \frac{1}{\sqrt{2}} \sum_{z_i} (-1)^{x_i z_i}|z_i\rangle = \frac{1}{\sqrt{2^n}} \sum_{z} (-1)^{x\cdot z}|z\rangle. \qquad \square$$

The Hadamard transform requires only n single-qubit gates (one H per qubit), compared to $O(n \cdot 2^n)$ operations classically. This exponential compression is possible because it acts on amplitudes in superposition, not on 2^n data entries individually. It is the prototype for the *Quantum Fourier Transform* in Section 2.

The key structural difference from classical computation is this: a classical algorithm that applies the Hadamard transform to a vector $v \in \mathbb{C}^{2^n}$ must read all 2^n entries of v. The quantum algorithm applies $\mathsf{H}^{\otimes n}$ to a quantum state in $O(n)$ gates, but the result is a *superposition*, not a list of coefficients. Extracting individual coefficients still requires measurement, which collapses the superposition. This tension between the exponential state space and the limited classical information one can extract from it is the real challenge in designing useful quantum algorithms.

Entanglement

Definition 1.6 (Entanglement). A state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is *separable* if $|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$ for some $|\phi\rangle \in \mathcal{H}_A$, $|\chi\rangle \in \mathcal{H}_B$; otherwise it is *entangled*.

The four *Bell states* form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$, all maximally entangled:

$$|\Phi^\pm\rangle = \tfrac{1}{\sqrt{2}}\big(|00\rangle \pm |11\rangle\big), \qquad |\Psi^\pm\rangle = \tfrac{1}{\sqrt{2}}\big(|01\rangle \pm |10\rangle\big).$$

Proposition 1.7 (Bell state preparation). $\mathsf{CNOT} \cdot (\mathsf{H} \otimes \mathbb{I})|00\rangle = |\Phi^+\rangle$.

Exercise 1.8. Verify Proposition 1.7. Then construct circuits (sequences of H, X, and CNOT gates) that prepare each of $|\Phi^-\rangle$, $|\Psi^+\rangle$, and $|\Psi^-\rangle$ from the input $|00\rangle$.

Remark 1.9. A separable state $|\phi\rangle \otimes |\chi\rangle$ corresponds to a rank-1 tensor in $\mathbb{C}^{2^m} \otimes \mathbb{C}^{2^n}$. When a general state is viewed as a matrix $M \in \mathbb{C}^{2^m \times 2^n}$, with M_{ij} the coefficient of $|ij\rangle$, the state is separable if and only if $\operatorname{rank}(M) = 1$. The singular value decomposition of M yields the *Schmidt decomposition*, and the number of non-zero singular values is the *Schmidt rank*, which quantifies the degree of entanglement.

Exercise 1.10. (a) Show that $|\Phi^+\rangle$ has Schmidt rank 2.

(b) For the state $|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$, find the Schmidt decomposition and determine whether it is entangled.

The following is a very important physical restriction in quantum computing. It is also one of the main reasons quantum computing is difficult.

Proposition 1.11 (No-cloning theorem [1]). *There is no unitary U satisfying*

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \qquad \textit{for all } |\psi\rangle \in \mathbb{C}^2.$$

Proof. Suppose such U exists. For any $|\phi\rangle, |\psi\rangle$, unitarity gives

$$\langle \phi \,|\, \psi \rangle^2 = (\langle \phi| \otimes \langle 0|)(U^\dagger U)(|0\rangle \otimes |\psi\rangle) = \langle \phi \,|\, \psi \rangle$$

so $\langle \phi \,|\, \psi \rangle \in \{0, 1\}$ for all pairs. This contradicts the existence of non-orthogonal, non-identical states such as $|0\rangle$ and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. $\square$

Reversible computation and oracles

Unitaries are invertible ($U^{-1} = U^\dagger$), so quantum computation is *reversible*. On the other hand, an irreversible classical map must be embedded reversibly before it can be run. A way to do this is the following: For $f : \{0,1\}^n \to \{0,1\}^m$ the standard embedding is the *(XOR) oracle*

$$U_f\,|x\rangle|y\rangle = |x\rangle\,|y \oplus f(x)\rangle, \qquad x \in \{0,1\}^n,\ y \in \{0,1\}^m,$$

which gives a permutation of basis states. Hence, U_f is unitary on $\mathbb{C}^n \otimes \mathbb{C}^m$ and is an involution ($U_f^2 = \mathbb{I}$). The input register x is preserved; the answer is XORed into the second register. Recall that $\oplus$ is the sum modulo 2, i.e., the XOR gate is a classical logic gate that outputs 0 when the components of y and $f(x)$ are equal, and 1 otherwise.

Phase kickback. Setting the target register to the X-eigenstate $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ turns the XOR oracle into a *phase oracle*. Indeed, since $\mathsf{X}|-\rangle = -|-\rangle$, it holds that

$$U_f\,|x\rangle|-\rangle = |x\rangle\,\frac{1}{\sqrt{2}}\big(|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle\big) = (-1)^{f(x)}\,|x\rangle|-\rangle.$$

The function value is “kicked back” onto the input register as a relative phase $(-1)^{f(x)}$, while $|-\rangle$ is left untouched. More generally, a controlled-U whose target is an eigenstate $|u\rangle$, e.g., $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ imprints the eigenphase on the control:

$$\text{c-}U\,|1\rangle|u\rangle = e^{2\pi i\varphi}\,|1\rangle|u\rangle.$$

Converting eigenvalue information into a measurable relative phase is the single mechanism behind quantum phase estimation (Section **??**) and, hence, behind every algorithm on the QPE spine, such as Grover (Section **??**).

2 Quantum Fourier Transform

The *Quantum Fourier Transform* is the computational engine of the algorithms in this course. It is the DFT matrix acting on amplitudes of a quantum state, and it admits an efficient $O(n^2)$-gate circuit arising from a product formula.

Why the Fourier transform? The DFT is ubiquitous in numerical analysis: it converts between a function and its frequency representation, diagonalizes circulant matrices, and accelerates convolution from $O(N^2)$ to $O(N \log N)$ via the FFT. In quantum computing, the DFT plays an analogous role: it converts between a computational-basis representation and a phase-encoded representation, and it is the tool by which quantum algorithms *read off* periodicity and spectral information from a quantum state. The key difference is that the QFT operates on the *amplitudes* of a superposition — it is applied to 2^n complex numbers simultaneously in $O(n^2)$ gate operations, a task that would take $O(n \cdot 2^n)$ classical operations. As with the Hadamard transform, this compression is only useful when the circuit's output state is processed by a subsequent algorithm (such as QPE) that extracts the relevant information without full tomography.

Definition and unitarity

Throughout this section, $n \in \mathbb{N}$ denotes the number of qubits and $N = 2^n$. We use three related but distinct objects:

- an *integer* $k \in \{0, \ldots, N-1\}$, written in decimal as usual;
- its *binary representation* $k = k_1 2^{n-1} + k_2 2^{n-2} + \cdots + k_n 2^0$, written as the bit string $[k_1 k_2 \cdots k_n]$ with each $k_j \in \{0, 1\}$;
- the *binary fraction* $[0.k_l k_{l+1} \cdots k_m] = k_l 2^{-1} + \cdots + k_m 2^{-(m-l+1)} \in [0, 1)$.

The computational basis state $|k\rangle$ is indexed by the *integer* $k \in \mathbb{N}$; we write $|k_1 k_2 \cdots k_n\rangle$ to mean the tensor product $|k_1\rangle \otimes |k_2\rangle \otimes \cdots \otimes |k_n\rangle$, which equals $|k\rangle$ when $k = \sum_j k_j 2^{n-j}$. Binary fractions will commonly appear in exponents as phases, e.g., $e^{2\pi i\,[0.k_1 k_2]}$.

The *Discrete Fourier Transform* (DFT) on $\mathbb{C}^N$ is the linear map F_N defined by

$$\hat{a}_k := (\mathsf{F}_N a)_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega_N^{jk}\, a_j, \qquad k = 0, \ldots, N-1, \qquad \omega_N := e^{2\pi i/N}.$$

with matrix elements $(\mathsf{F}_N)_{kj} = \omega_N^{jk}/\sqrt{N}$, $j, k = 0, \ldots, N-1$.

Lemma 2.1 (Unitarity of the DFT). $\mathsf{F}_N^\dagger \mathsf{F}_N = \mathbb{I}_N$.

Proof. $(\mathsf{F}_N^\dagger \mathsf{F}_N)_{lj} = \frac{1}{N} \sum_{k=0}^{N-1} \omega_N^{k(j-l)}$. For $j = l$ this equals 1. For $j \neq l$, the sum is a geometric series with ratio $\omega_N^{j-l} \neq 1$, which vanishes. $\square$

Definition 2.2 (Quantum Fourier Transform). The *n-qubit Quantum Fourier Transform* QFT_N $(N = 2^n)$ is the unitary operator defined on the computational basis by

$$\mathsf{QFT}_N |k\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega_N^{jk} |j\rangle, \qquad k = 0, \ldots, N-1.$$

As a matrix, $\mathsf{QFT}_N = \mathsf{F}_N$, which is unitary by Lemma 2.1.

Remark 2.3. The QFT is the DFT matrix F_N applied to the *amplitude vector* of a quantum state rather than to a classical data vector. A classical computer requires $O(N \log N)$ arithmetic operations (FFT) to apply F_N to an N-vector. The QFT circuit uses only $O(n^2) = O((\log N)^2)$ gates, but it acts on a quantum superposition of all N basis states simultaneously. This apparent compression is why the QFT is the engine of Shor's algorithm and QPE, though extracting the result requires careful design of the surrounding algorithm.

From the definition of the QFT, it is not apparent that it is a transformation on n qubits. The following statement makes it explicit.

Theorem 2.4 (Product formula). *The quantum Fourier transform takes the form*

$$\mathsf{QFT}_N|k_1k_2\cdots k_n\rangle = \bigotimes_{l=1}^{n} \frac{1}{\sqrt{2}}\Big(|0\rangle + e^{2\pi i\,[0.k_{n-l+1}\cdots k_n]}|1\rangle\Big).$$

Proof. Substituting the binary expansion of k and $|j\rangle = |j_1\rangle \otimes \cdots \otimes |j_n\rangle$:

$$\begin{aligned}
\mathsf{QFT}_N|k\rangle &= \frac{1}{\sqrt{N}} \sum_{j_1,\ldots,j_n\in\{0,1\}} e^{2\pi i\,k\sum_{l=1}^{n} j_l 2^{-l}}|j_1\rangle \otimes \cdots \otimes |j_n\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{j_1,\ldots,j_n\in\{0,1\}} \prod_{l=1}^{n} e^{2\pi i\,k\cdot 2^{-l}j_l}|j_1\rangle \otimes \cdots \otimes |j_n\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{j_1,\ldots,j_n\in\{0,1\}} \big(e^{2\pi i\,k\cdot 2^{-l}j_1}|j_1\rangle\big) \otimes \cdots \otimes \big(e^{2\pi i\,k\cdot 2^{-l}j_n}|j_n\rangle\big) \\
&= \frac{1}{\sqrt{N}} \bigotimes_{l=1}^{n}\Big(|0\rangle + e^{2\pi i\,k\cdot 2^{-l}}|1\rangle\Big).
\end{aligned}$$

Writing $k = k_1 2^{n-1} + \cdots + k_n$, we have $k \cdot 2^{-l} = k_1 2^{n-1-l} + \cdots + k_{n-l} + [0.k_{n-l+1}\cdots k_n]$. The integer part contributes only multiples of 2π to the exponent, so $e^{2\pi i\,k\cdot 2^{-l}} = e^{2\pi i\,[0.k_{n-l+1}\cdots k_n]}$. $\square$

Remark 2.5. The product formula reveals two key structural facts. First, the output is a *tensor product* of single-qubit states, enabling a qubit-by-qubit circuit. Second, the l-th factor depends only on the last l bits $k_{n-l+1}, \ldots, k_n$ of the input, so the output qubits are in *reversed* order relative to the input.

Exercise 2.6. Show that QFT_N diagonalizes the cyclic shift operator $\widehat{\mathsf{X}}|k\rangle = |k+1 \bmod N\rangle$, i.e, compute the eigenvectors and eigenvalues of $\widehat{\mathsf{X}}$, and verify that

$$\mathsf{QFT}_N^\dagger\,\widehat{\mathsf{X}}\,\mathsf{QFT}_N = \widehat{\mathsf{Z}}^{-1},$$

where $\widehat{\mathsf{Z}}|k\rangle = \omega_N^k|k\rangle$.

Interpret this as the quantum analog of the DFT diagonalizing the derivative operator.

Exercise 2.7. Show that $\mathsf{QFT}_N^4 = \mathbb{I}$ for all $N \in \mathbb{N}$, and deduce that all the eigenvalues of QFT_N lie in the set $\{1, i, -1, -i\}$.

Exercise 2.8. Apply QFT_4 to the states $|1\rangle$ and $|2\rangle$ using the product formula (cf. Theorem 2.4). Express each result as a tensor product of single-qubit states and verify using the definition that

$$\mathsf{QFT}_4|k\rangle = \frac{1}{2}\sum_{j=0}^{3} i^{jk}|j\rangle.$$

2.1 The QFT circuit

The product formula directly yields an efficient circuit. Define the *phase rotation gate*

$$\mathsf{R}_m = \operatorname{diag}\bigl(1, e^{2\pi i/2^m}\bigr), \quad m \in \mathbb{N},$$

and the *controlled*-R_m gate, which applies R_m to the target iff the control is $|1\rangle$, i.e.,

$$\text{c-}\mathsf{R}_m = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes \mathsf{R}_m.$$

Theorem 2.9 (QFT circuit complexity [1]). QFT_N *can be implemented using* n *Hadamard gates,* $\binom{n}{2}$ *controlled-*R_m *gates, and* $\lfloor n/2 \rfloor$ SWAP *gates. Total gate count:* $\frac{n(n+1)}{2} + \lfloor n/2 \rfloor = O(n^2)$.

Example 2.10 ($n = 3$ qubits). In Figure 2.1, we trace the circuit for input $|k\rangle = |k_1 k_2 k_3\rangle$.

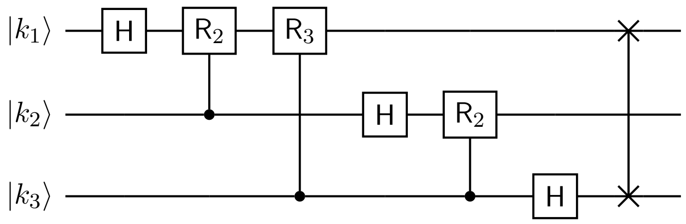

Figure 2.1: QFT circuit for $n = 3$ qubits. After the final SWAP the output order matches the product formula.

We begin by tracing what happens to qubit 1 (the most significant):

(1) **H on qubit 1:** $|k_1\rangle \to \frac{1}{\sqrt{2}}\bigl(|0\rangle + (-1)^{k_1}|1\rangle\bigr) = \frac{1}{\sqrt{2}}\bigl(|0\rangle + e^{2\pi i\,[0.k_1]}|1\rangle\bigr)$.

(2) **Controlled-R_2 (control: qubit 2):** $\to \frac{1}{\sqrt{2}}\bigl(|0\rangle + e^{2\pi i\,[0.k_1 k_2]}|1\rangle\bigr)$.

$$|0\rangle \frac{1}{\sqrt{2}}\bigl(|0\rangle + e^{2\pi i\,[0.k_1]}|1\rangle\bigr) \to |0\rangle \frac{1}{\sqrt{2}}\bigl(|0\rangle + e^{2\pi i\,(k_1 2^{-1} + k_2 2^{-2})}|1\rangle\bigr)$$
$$|1\rangle \frac{1}{\sqrt{2}}\bigl(|0\rangle + e^{2\pi i\,[0.k_1]}|1\rangle\bigr) \to |1\rangle \frac{1}{\sqrt{2}}\bigl(|0\rangle + e^{2\pi i\,(k_1 2^{-1} + k_2 2^{-2})}|1\rangle\bigr)$$

(3) **Controlled-R_3 (control: qubit 3):** $\to \frac{1}{\sqrt{2}}\bigl(|0\rangle + e^{2\pi i\,[0.k_1 k_2 k_3]}|1\rangle\bigr)$.

Each controlled-R_m appends one further binary digit to the phase. The same procedure on qubit 2 yields $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\,[0.k_2 k_3]}|1\rangle)$, and H alone on qubit 3 gives $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\,[0.k_3]}|1\rangle)$. After the SWAP, the output matches Theorem 2.4. Gate count for $n = 3$: $3 + 3 + 1 = 7$, consistent with $\frac{3\cdot 4}{2} + 1 = 7$. For general n: qubit l receives H plus $(n - l)$ controlled rotations, giving $\sum_{l=1}^{n}(n - l) = \binom{n}{2}$ controlled-R_m gates, n Hadamards, and $\lfloor n/2 \rfloor$ SWAPs. $\square$

Since QFT_N is unitary, $\mathsf{QFT}_N^{-1} = \mathsf{QFT}_N^{\dagger}$. As usual, the inverse QFT serves as a basis change, transforming from the Fourier basis to the computational basis.

Proposition 2.11 (Inverse QFT circuit). $\mathsf{QFT}_N^{\dagger}$ *is implemented by reversing the QFT circuit and replacing each* R_m *with* $\mathsf{R}_m^{\dagger} = \mathsf{R}_{-m}$ *(phase* $e^{-2\pi i/2^m}$*).*

Remark 2.12. The gates R_m apply exponentially small rotations $e^{2\pi i/2^m}$ for $m \gg 1$ large and may be omitted in practice; see Section 2.2.

2.2 Approximate QFT

Definition 2.13 (Approximate QFT). The *approximate QFT* with cutoff $m_{\max}$ is obtained by dropping all controlled-R_m gates with $m > m_{\max}$. It uses $\Theta(n\, m_{\max})$ gates instead of $\Theta(n^2)$.

Theorem 2.14 (Approximation error). *Let U and U_{approx} be the exact and approximate QFT unitaries. Then*

$$\|U - U_{\text{approx}}\|_{op} \leq \frac{2\pi n}{2^{m_{\max}}},$$

where $\|\cdot\|_{op}$ denotes the operator norm. Consequently, choosing $m_{\max} = \lceil \log_2(n/\varepsilon) \rceil$ yields an error less than $\varepsilon > 0$ with only $O(n \log(n/\varepsilon))$ gates.

Proof. Write $U = G_1 G_2 \cdots G_K$ and $U_{\text{approx}} = G'_1 \cdots G'_K$, where $G'_i = G_i$ for kept gates and $G'_i = \mathbb{I}$ for dropped gates. By the triangle inequality applied inductively to products of unitaries,

$$\|U - U_{\text{approx}}\|_{\text{op}} \leq \sum_{i:\, G_i \text{ dropped}} \|G_i - \mathbb{I}\|_{\text{op}}.$$

A c-R_m gate in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ is $\operatorname{diag}(1, 1, 1, e^{2\pi i/2^m})$. Hence,

$$\|\text{c-}\mathsf{R}_m - \mathbb{I}\|_{\text{op}} = |e^{2\pi i/2^m} - 1| = 2\sin\left(\frac{\pi}{2^m}\right) \leq \frac{2\pi}{2^m},$$

where we used the estimates $|e^{i\theta} - 1| = 2|\sin(\theta/2)|$ and $|\sin\theta| \leq |\theta|$. Summing over all dropped gates (those with $m > m_{\max}$):

$$\|U - U_{\text{approx}}\|_{\text{op}} \leq \sum_{l=1}^{n} \sum_{m=m_{\max}+1}^{n-l+1} \frac{2\pi}{2^m} \leq n \sum_{m=m_{\max}+1}^{\infty} \frac{2\pi}{2^m} = \frac{2\pi n}{2^{m_{\max}}}.$$ □

Remark 2.15. The approximation strategy mirrors two classical ideas. First, it is analogous to a *low-pass filter*: rotations R_m with $m > m_{\max}$ correspond to high-frequency components $e^{2\pi i/2^m}$ near 1, which contribute negligibly to the output and can be discarded. Second, the gate count $O(n \log(n/\varepsilon))$ matches the complexity of computing a *sparse* DFT (keeping only the $O(n\, m_{\max})$ most significant frequency interactions), and is reminiscent of the complexity of quadrature rules with adaptive frequency truncation. The Frobenius error bound plays the role of the aliasing error estimate in such truncated transforms.

Exercise 2.16. For $n = 4$ qubits and cutoff $m_{\max} = 2$, identify which controlled-R_m gates are dropped. Compute the Frobenius error bound from Theorem 2.14 and compare with the total number of gates saved.